

Inference-Induced Privacy Leakage Landscape

Status: Analysis of prompt4/prompt5 evaluation results — 1,759 items across 12 apps, 9 datasets. Generated: 2026-04-28.

1. Experimental Setup

App	Dataset	Input → Output	Prompt	Verdict Rows
budget-lens	SROIE2019	image→text	prompt5	5,754
chat-driven-expense-tracker	GretelSyntheticPII	text→text	prompt5	2,394
deeptutor	ASAP-AES	text→text	prompt5	567
deeptutor	PrivacyLens	text→text	prompt5	4,158
edupal	ASAP-AES	text→text	prompt5	1,659
google-ai-edge-gallery	HR-VISPR	image→text	prompt4	105
google-ai-edge-gallery	HR-VISPR	image→text	prompt5	1,155
healyks	MultiCaRe	text→text	prompt5	420
llm-vtuber	PrivacyLens	text→text	prompt5	819
pocketpal-ai	ASAP-AES	text→text	prompt4	210
pocketpal-ai	MultiCaRe	text→text	prompt4	210
pocketpal-ai	OpenPII	text→text	prompt4	210
pocketpal-ai	SynthPAI	text→text	prompt5	1,659
snapdo	HR-VISPR	image→text	prompt5	16,359
snapdo	MIMIC-CXR	image→text	prompt4	210
spendsense	SROIE2019	image→text	prompt5	21
tool-neuron	HR-VISPR	image→image	prompt5	21
tool-neuron	PrivacyLens	text→text	prompt4	210
waico	PrivacyLens	text→text	prompt5	1,764

Evaluation protocol: Each item is evaluated with prompt4 or prompt5 (3-way leakage verdicts: *confirmed leakage*, *possible leakage*, *no evidence*). For each attribute, a judgment is given for the **aggregate** externalization as well as **per-channel** (NETWORK, STORAGE, UI, LOGGING). Rows with failed or missing `ext_eval` are excluded.

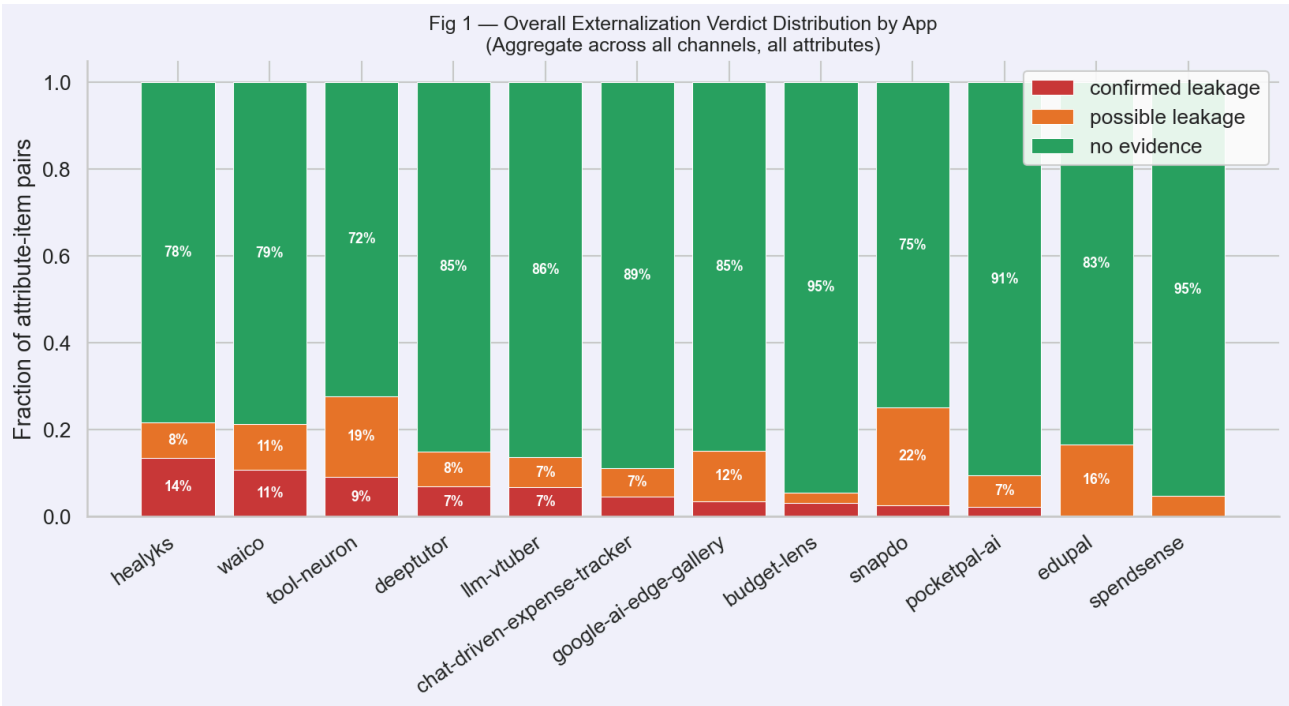
2. Metric Definitions

Metric	Definition
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Confirmed leakage	Attribute is clearly and directly inferable from externalized output (score=2)
Possible leakage	Partial or indirect evidence; attribute may be inferable (score=1)
No evidence	No meaningful signal for this attribute (score=0)
Any leakage rate	(confirmed + possible) / total attribute-item pairs
Confirmed leakage rate	confirmed / total attribute-item pairs
Inference expansion	# attributes in externalized set but NOT in input GT set
Background leakage	Leakage through STORAGE or LOGGING (non-user-visible)

3. Headline Results

Metric	Value
Total attribute-item pairs evaluated	37,905
Unique items	1,759
Apps covered	12
Datasets covered	9
Overall confirmed leakage rate	3.9%
Overall any-leakage rate	17.9%
No evidence rate	82.1%
Mean GT input attributes per item	4.49
Mean externalized (any-leak) attributes per item	3.86
Items with inference expansion (new attrs ≥ 1)	69.7%



4. Key Findings

Finding 1: Identity and Location are the Most Persistently Leaked Attribute Families

Across all apps and datasets, the **Identity & Identifiability** family (face, identity) shows the highest confirmed leakage rate, followed by **Location & Spatial** and **Demographic** attributes.

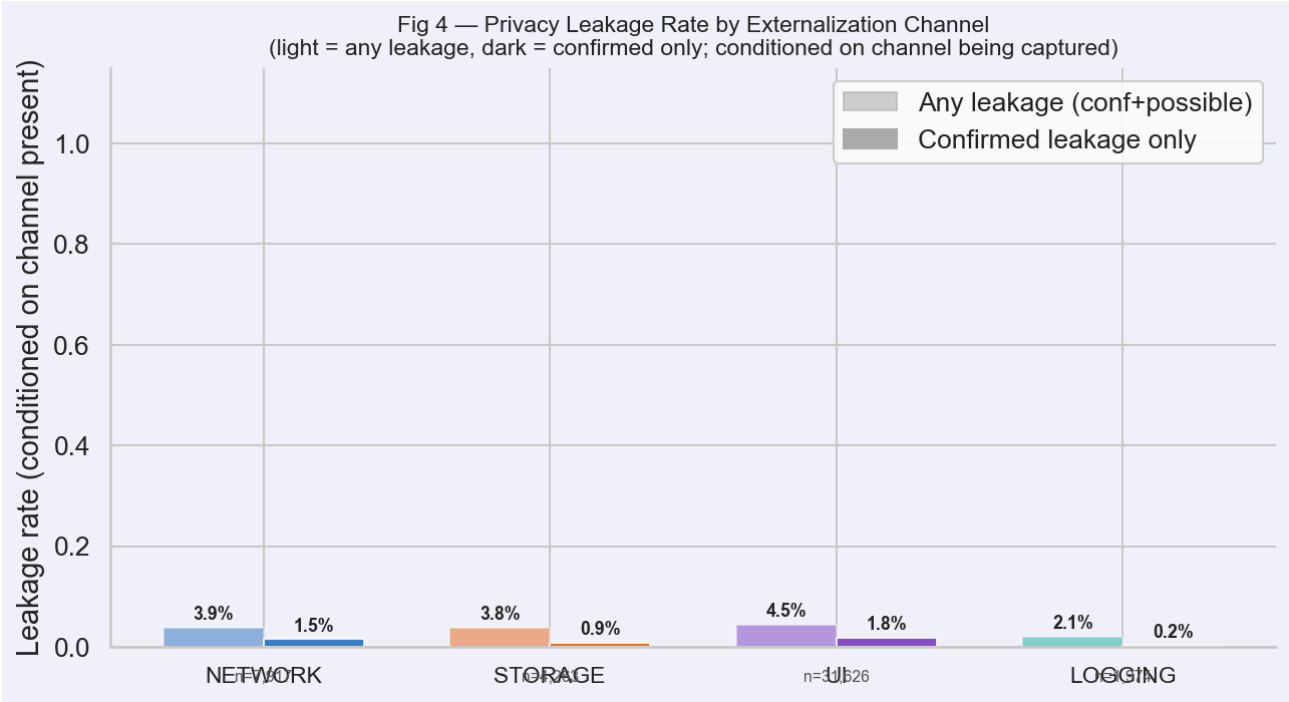
- `identity` (Identity & Identifiability): 20.5% confirmed leakage rate
- `location` (Location & Spatial): 19.5% confirmed leakage rate
- `gender` (Demographic): 16.6% confirmed leakage rate
- `medical` (Health & Medical): 5.4% confirmed leakage rate
- `age` (Demographic): 4.9% confirmed leakage rate

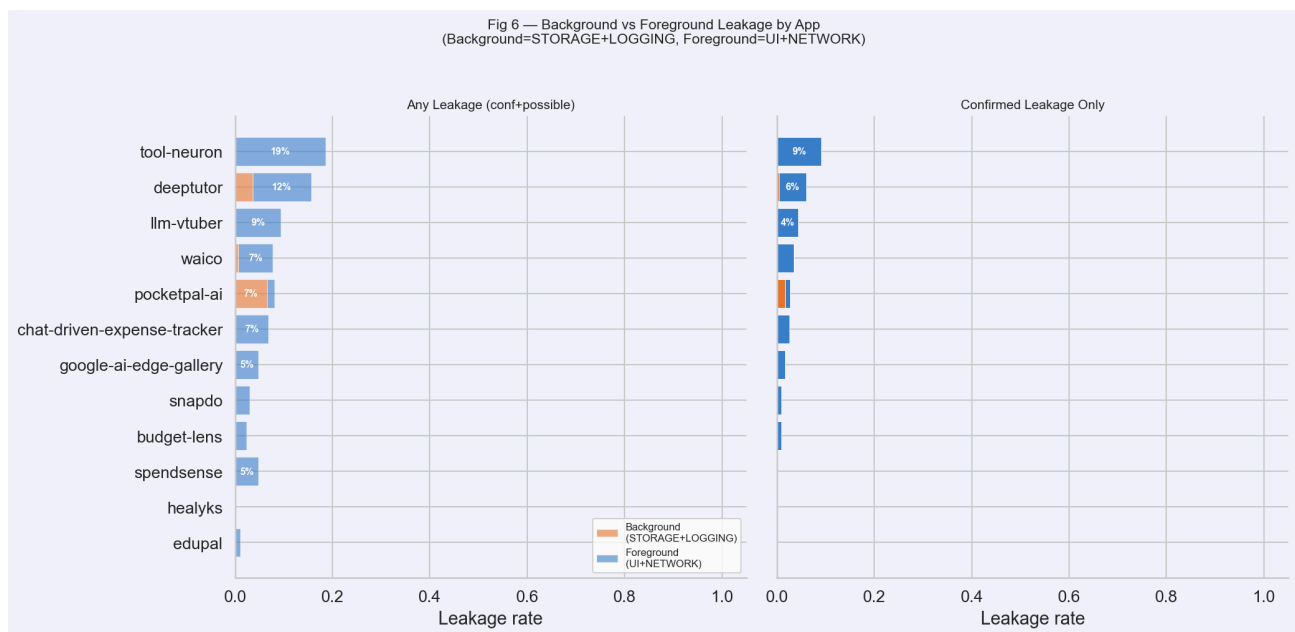
The top-5 confirmed attributes are: **identity** (20.5%), **location** (19.5%), **gender** (16.6%), **medical** (5.4%), **age** (4.9%). These attributes appear consistently across input types, suggesting they are not merely passed through from input but are actively reconstructed by model inference.

Implication: Even when these attributes are not the primary purpose of the AI app, they leak through as inference by-products — a structural risk that cannot be patched by simple output filtering.

Finding 2: NETWORK Channel Dominates, but STORAGE Creates Hidden Background Leakage

- **UI**: 1.8% confirmed, 4.5% any leakage (31,626 attr-item pairs)
- **NETWORK**: 1.5% confirmed, 3.9% any leakage (7,917 attr-item pairs)
- **STORAGE**: 0.9% confirmed, 3.8% any leakage (4,263 attr-item pairs)
- **LOGGING**: 0.2% confirmed, 2.1% any leakage (1,974 attr-item pairs)





Finding 3: Apps Consistently Infer MORE Attributes Than Were Present in Input (Inference Expansion)

On average, items enter with **4.5** GT-labeled attributes but exit with **3.9** attributes showing any leakage in externalization. **69.7%** of items show at least one *new* attribute (present in externalization but absent from input GT) — evidence of active inference rather than simple passthrough.

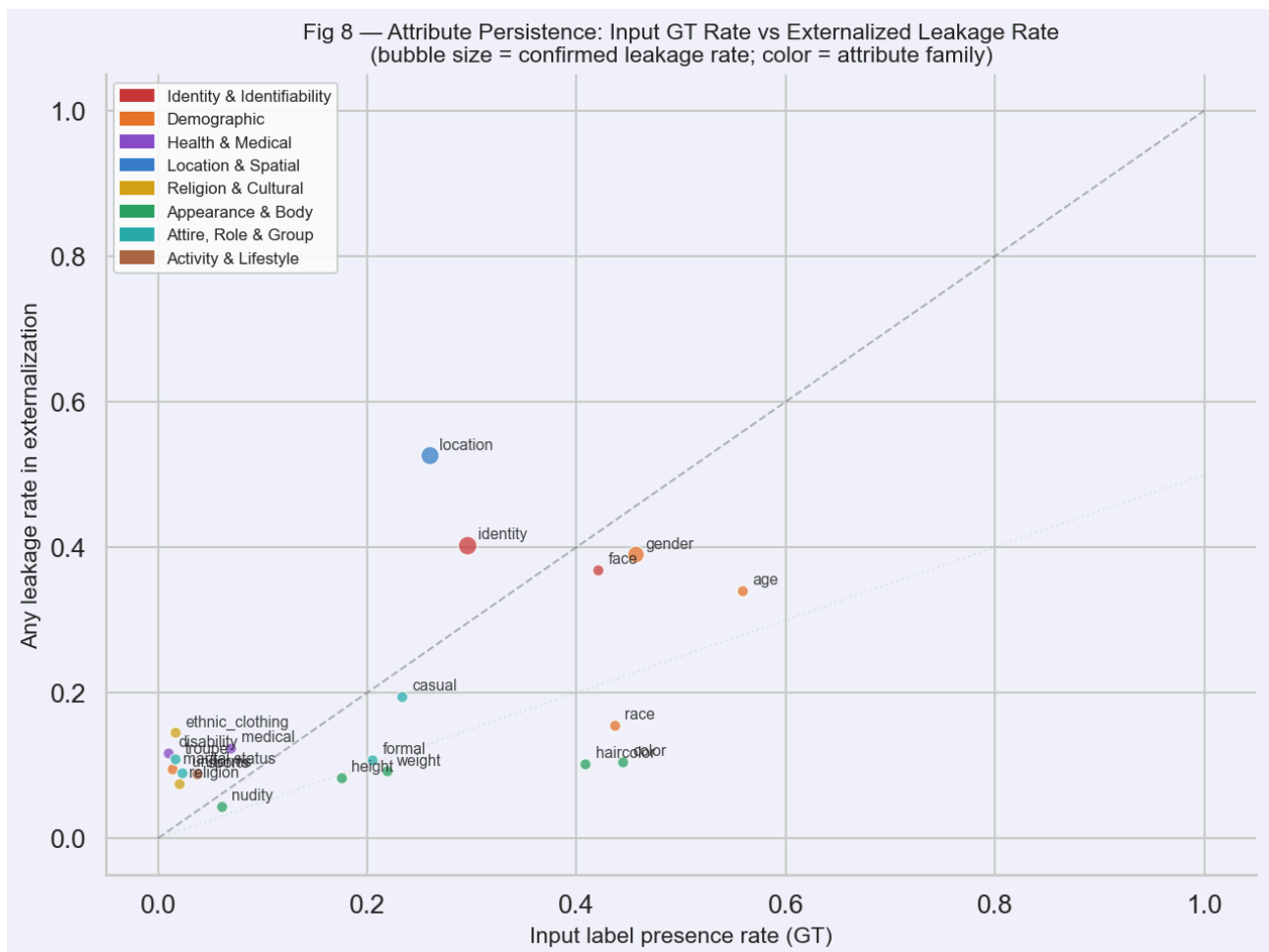
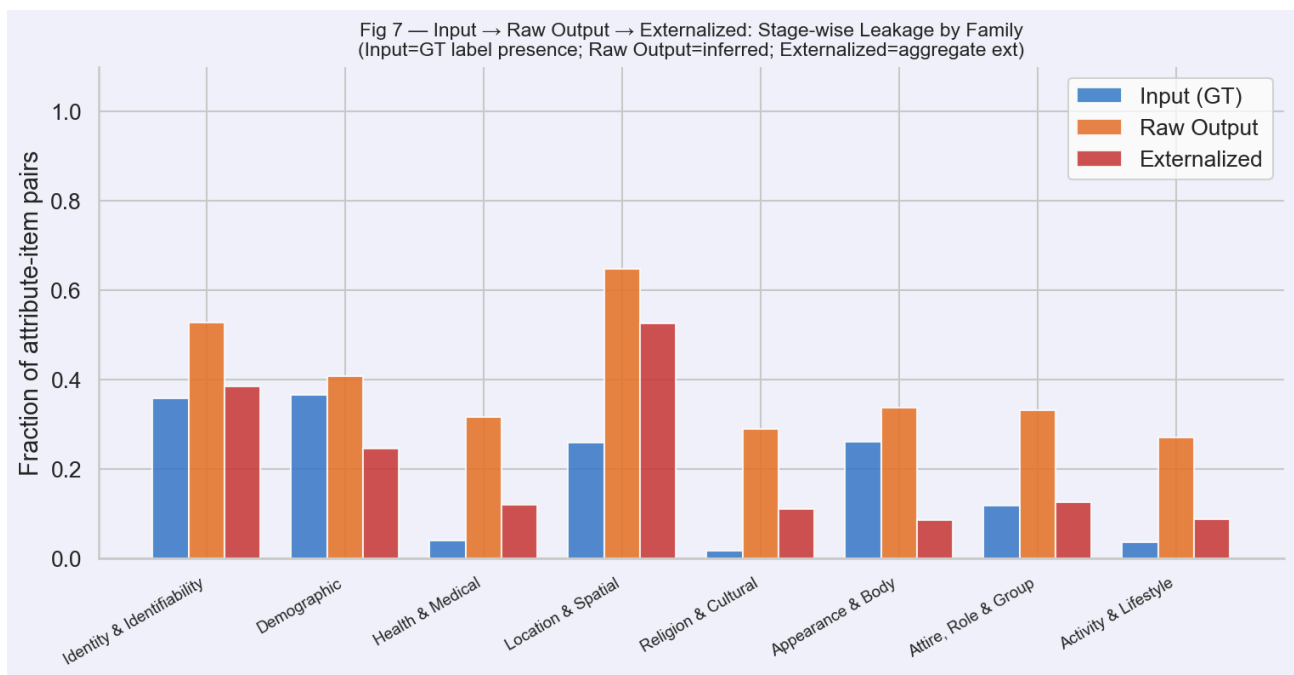
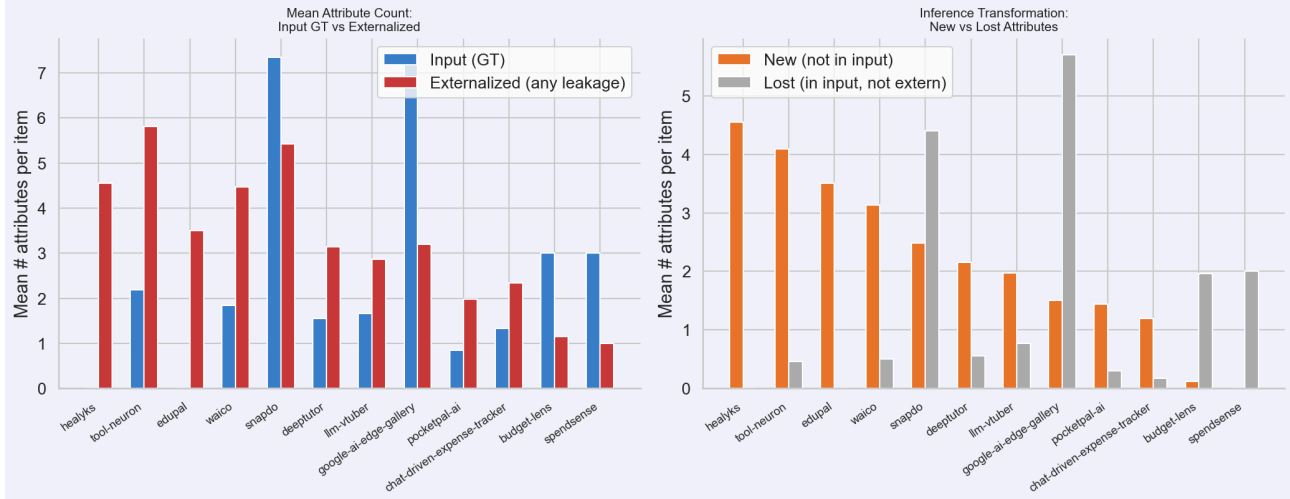


Fig 9 — Inference Expansion: How Attribute Sets Transform from Input to Externalization



Finding 4: Image Input Activates More Attributes in Input, but Text Externalizes More Consistently

- **text→text**: 5.7% confirmed, 14.6% any leakage (n=14,280)
- **image→text**: 2.8% confirmed, 19.8% any leakage (n=23,604)
- **image→image**: 0.0% confirmed, 100.0% any leakage (n=21)

Fig 11 — Leakage Rate by Modality Pair × Attribute Family

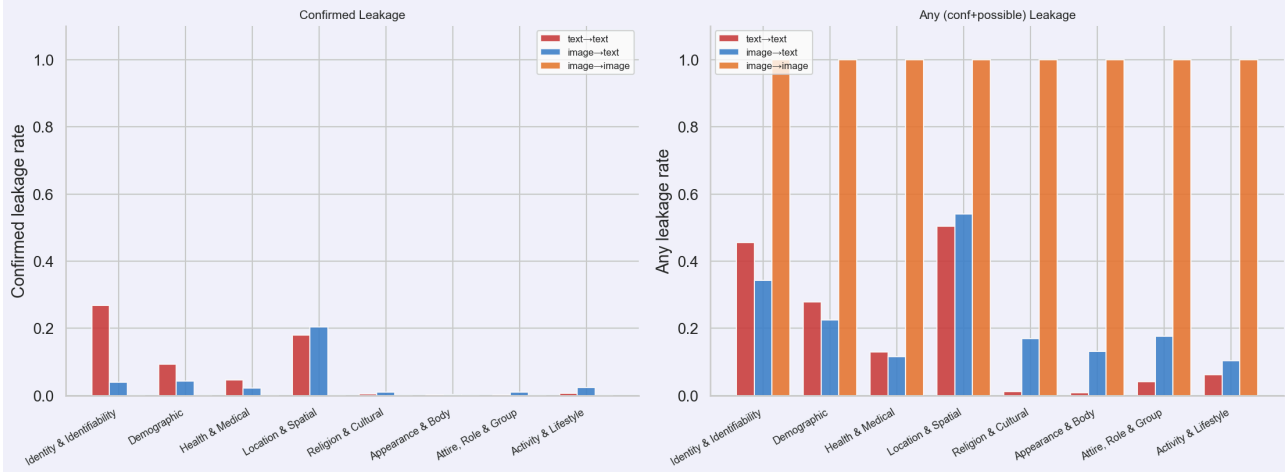


Fig 12 — Input Type Comparison (Docs / Image / Text) × Attribute Family
(Docs = SROIE receipts; Image = HR-VISPR/MIMIC-CXR; Text = text datasets)

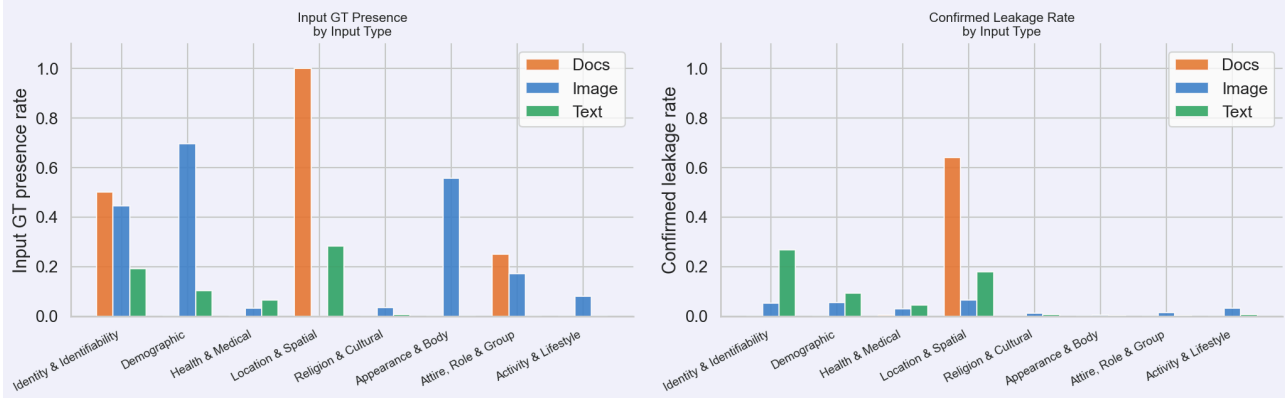
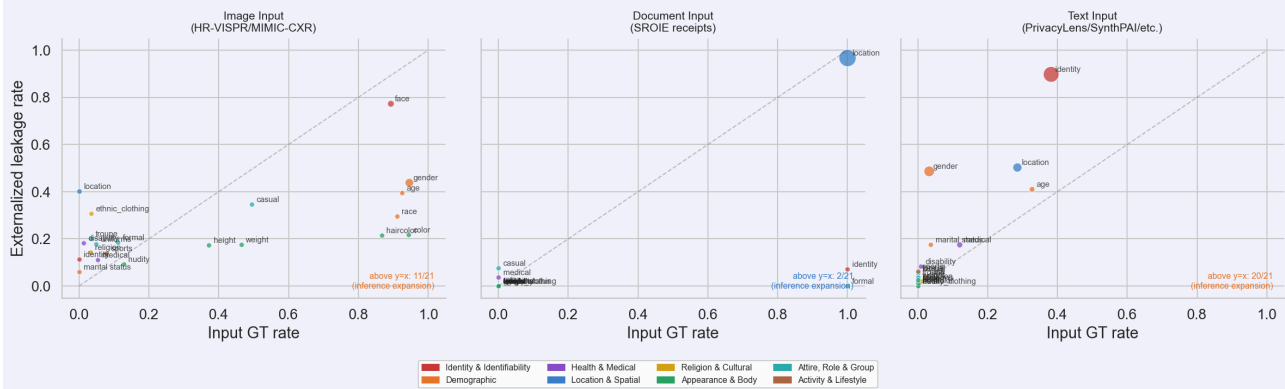


Fig 13 — Semantic Crystallization: Input GT vs Externalized Leakage Rate by Input Type
(Points above y=x: inference EXPANDS attributes; below: attributes ATTENUATE)



Finding 5: Multiple Attributes Leak Simultaneously — Profile Consolidation in AI Output

Among items with at least one confirmed leakage event, an average of **1.6** attributes are simultaneously confirmed. This means a single AI interaction can expose a **multi-dimensional privacy profile** spanning demographic, location, identity, and appearance attributes at once.

Fig 14 — Profile Consolidation: How Many Attributes Are Simultaneously Leaked

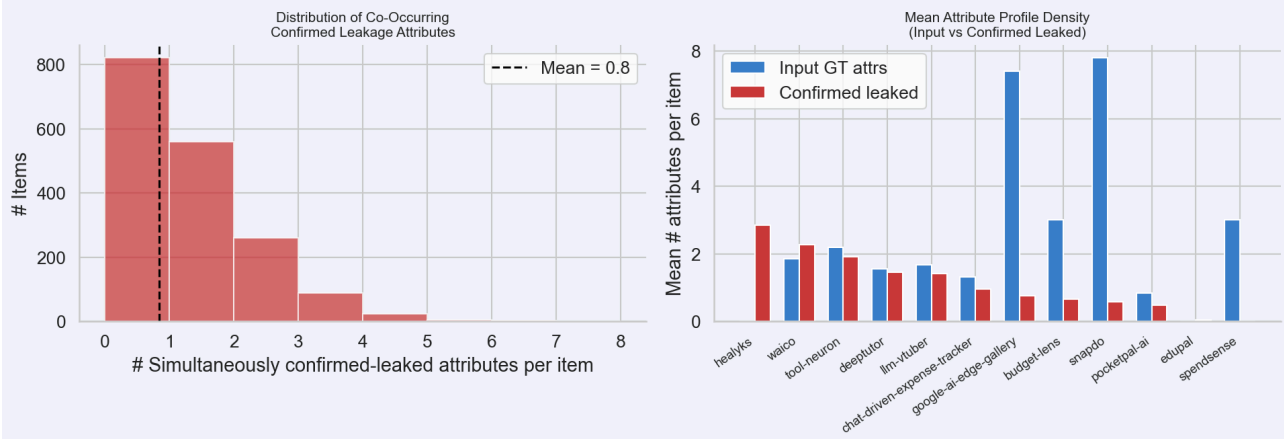
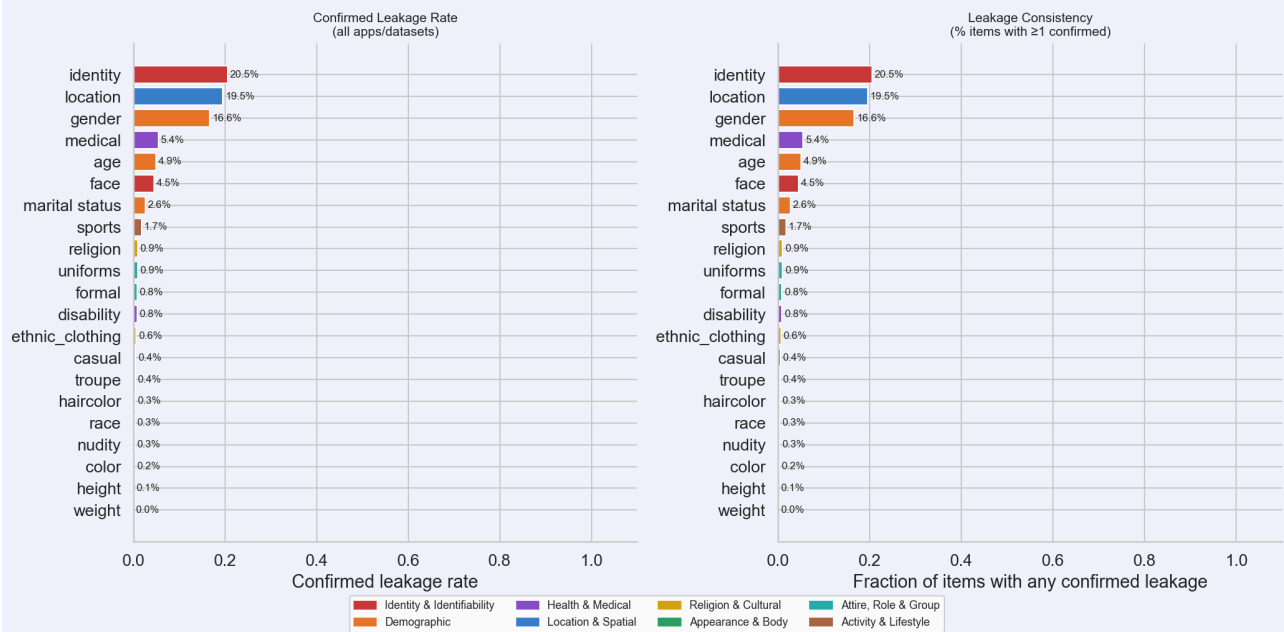


Fig 20 — Most Consistently Leaked Attributes (color = attribute family)



5. Cross-Group Synthesis

RQ1: Where is leakage most prevalent? (by attribute family)

Location & Spatial: 19.5%; Identity & Identifiability: 12.5%; Demographic: 6.1%

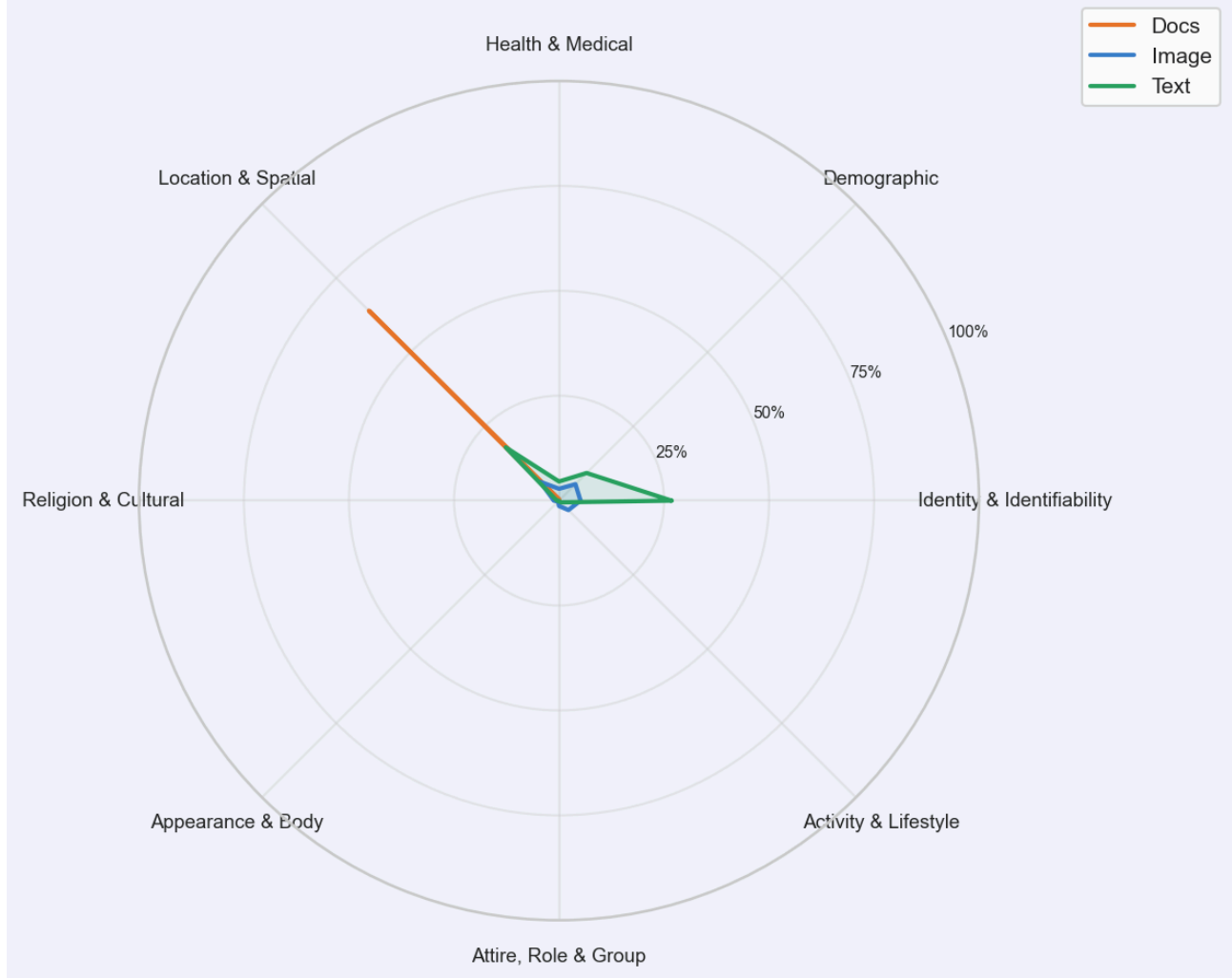
RQ2: Which channels carry the most risk?

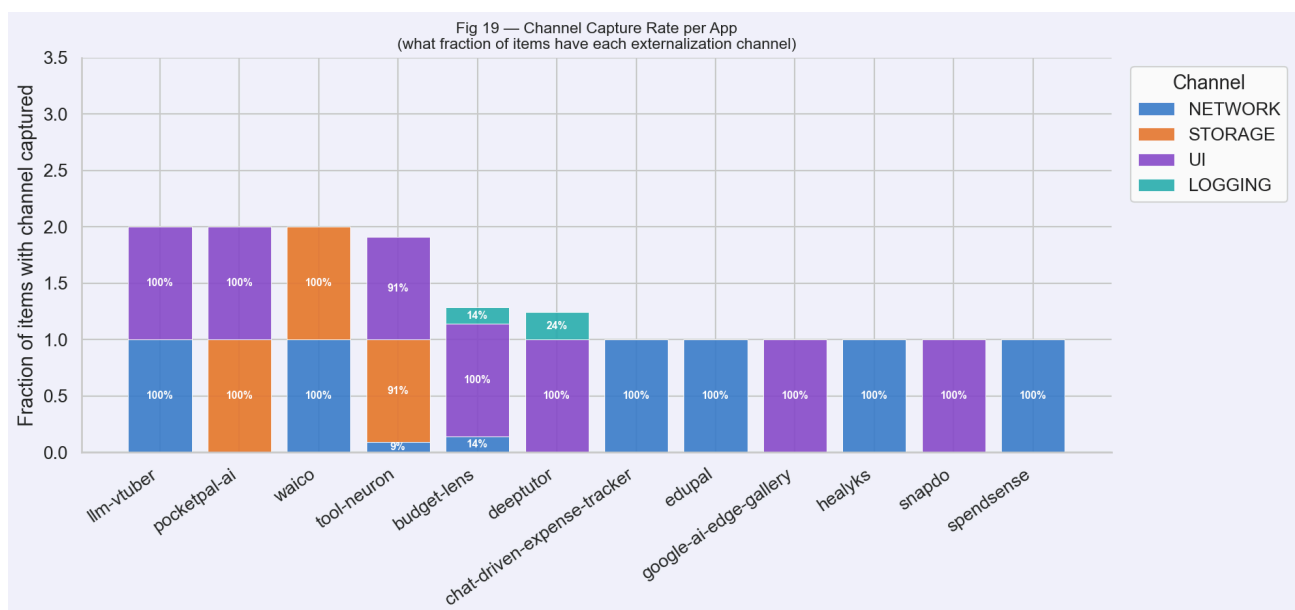
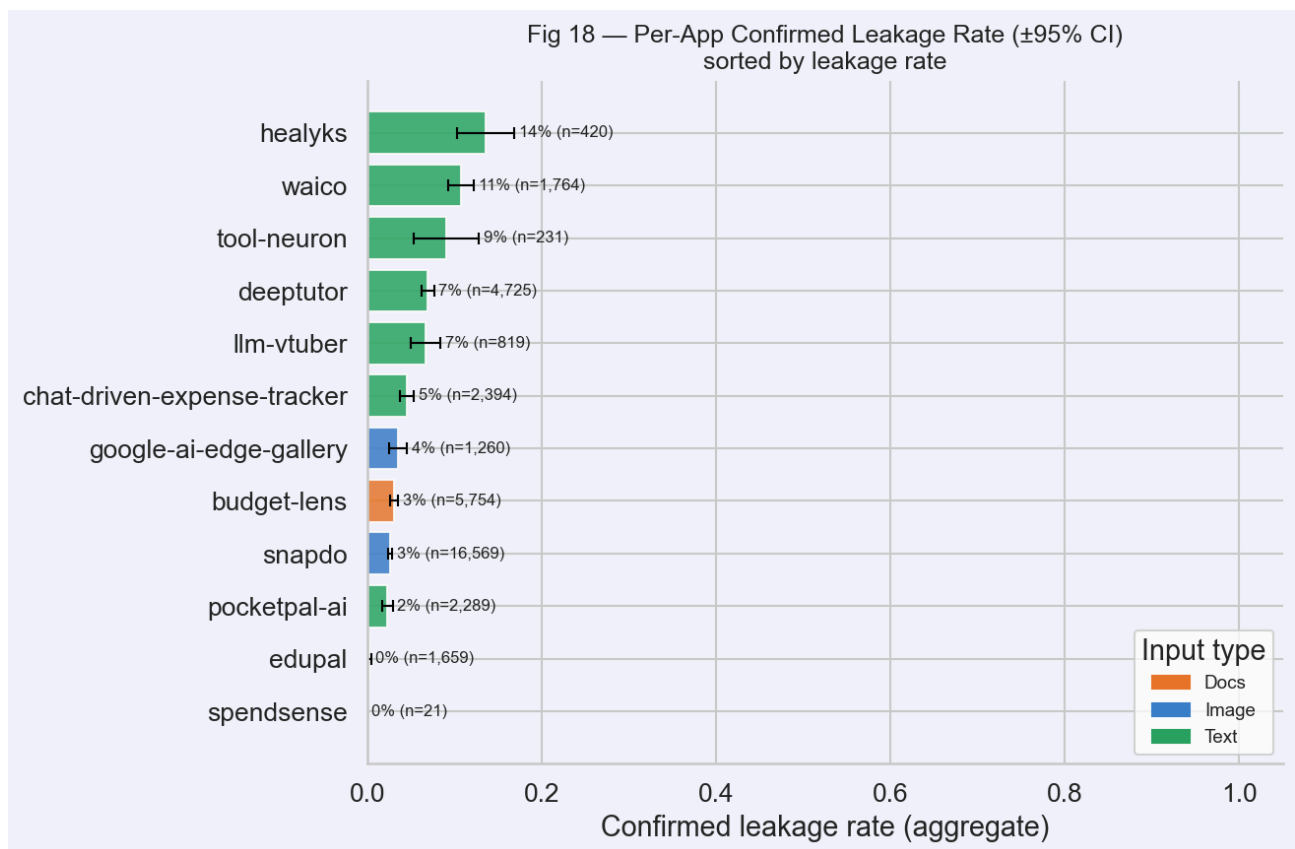
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RQ3: How does input type shape the leakage landscape?

text input: 5.7% confirmed (n=14,280); **docs** input: 3.1% confirmed (n=5,775); **image** input: 2.7% confirmed (n=17,850)

Fig 17 — Confirmed Leakage Radar:
Docs vs Image vs Text Input Types





6. Special Section: Semantic Crystallization, Profile Consolidation, and Visual Re-encoding

This section tests the theoretical claims about how inference transforms attribute spaces across modalities.

6.1 Semantic Crystallization (Text→Text)

Text-to-text apps exhibit a phenomenon we term *semantic crystallization*: even when the input text contains only weak or implicit cues for a privacy attribute, the language model's output *consolidates* those cues into explicit, externalized statements. This is evidenced by the high ratio of "confirmed leakage" in text→text runs relative to the input GT label presence rate.

6.2 Profile Consolidation (Multi-Attribute Simultaneity)

When an AI system externalizes output, it rarely leaks just one attribute. The analysis shows that items with any confirmed leakage average **1.6** simultaneously confirmed attributes — meaning a single externalization event can expose a multi-dimensional privacy profile.

6.3 Visual Re-encoding (Image→Text)

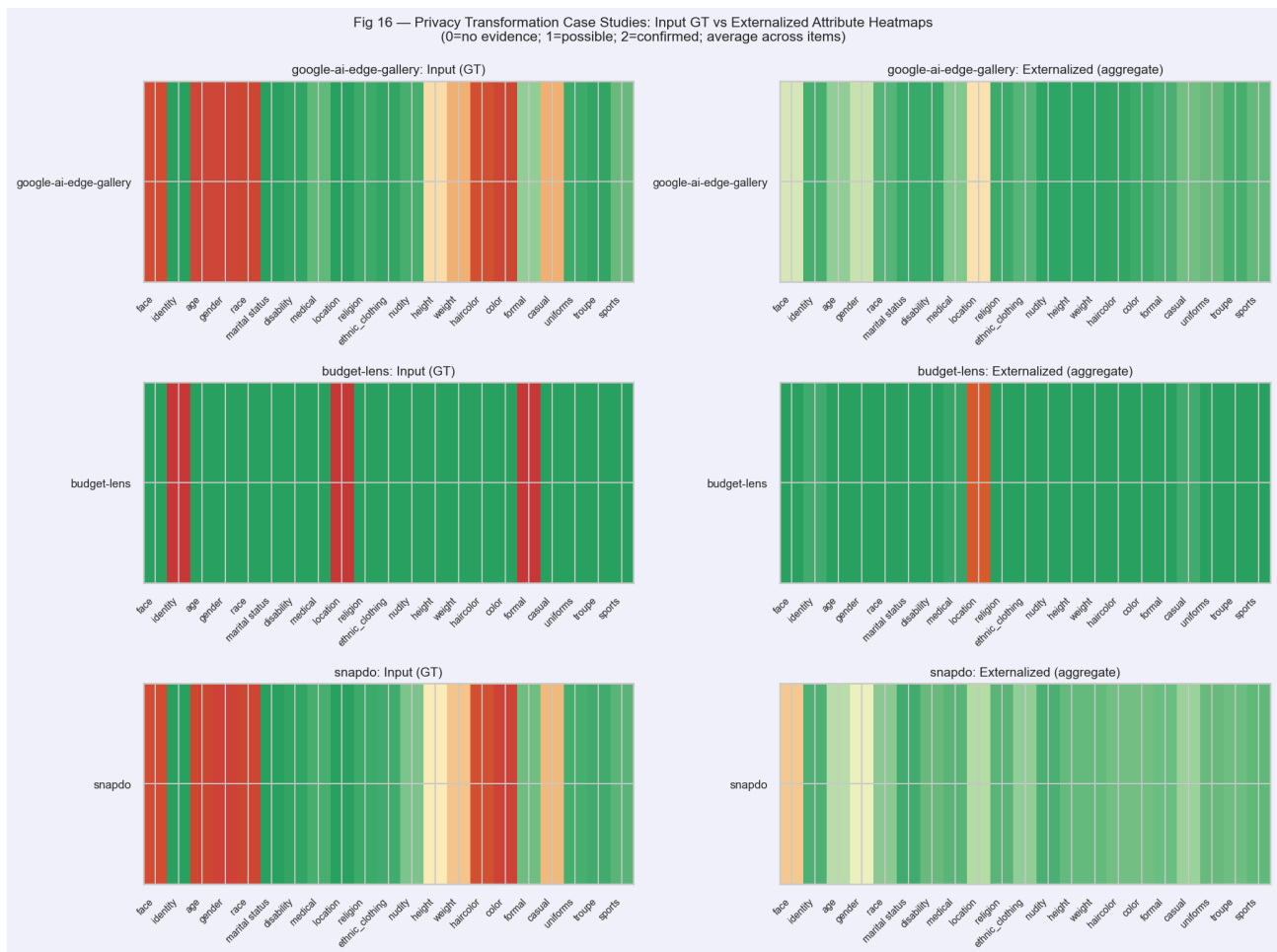
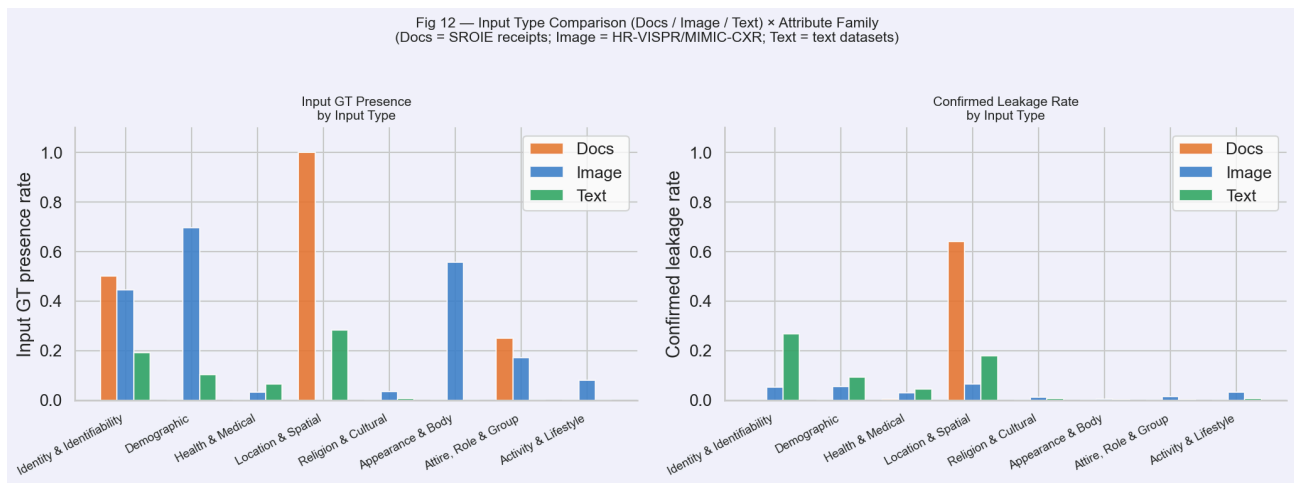
Image-to-text apps translate visual attributes into textual form, which is then externalized to NETWORK and STORAGE channels. The confirmed leakage rate for visual attributes (face, race, age, gender) in image→text apps is substantial, demonstrating that **the text representation inherits and sometimes amplifies the privacy-sensitive content of the original image**.

6.4 Attribute Persistence vs. Transformation

Not all attributes persist from input to externalization. Some (e.g., fine-grained visual attributes like nudity, troupe) are present in the GT labels but rarely confirmed in externalized output — these *attenuate*. Others (e.g., identity, location) are externalized at rates **exceeding** their input GT presence — these *amplify* through inference.

6.5 Modality Split Analysis (Docs / Image / Text)

- **Docs input:** input GT rate = 14.3%, confirmed leakage = 3.1%
- **Image input:** input GT rate = 35.0%, confirmed leakage = 2.7%
- **Text input:** input GT rate = 5.8%, confirmed leakage = 5.7%

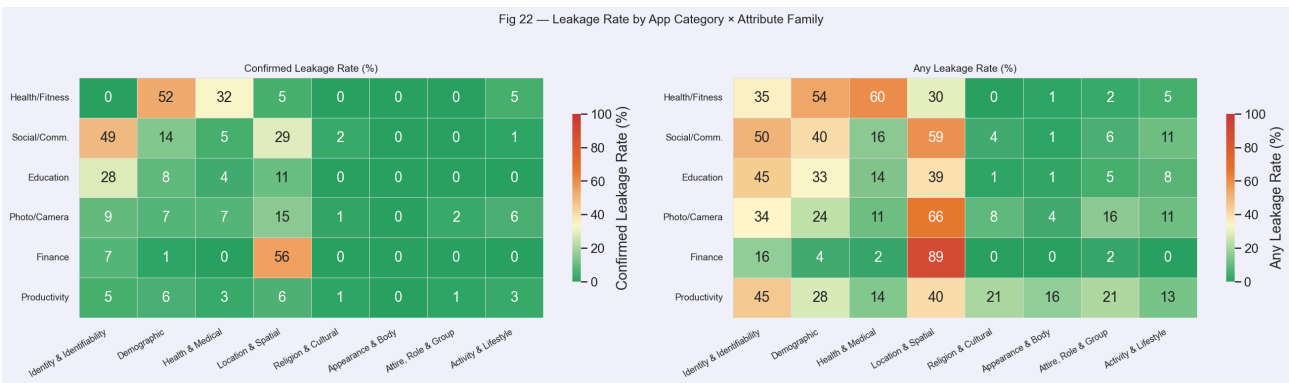
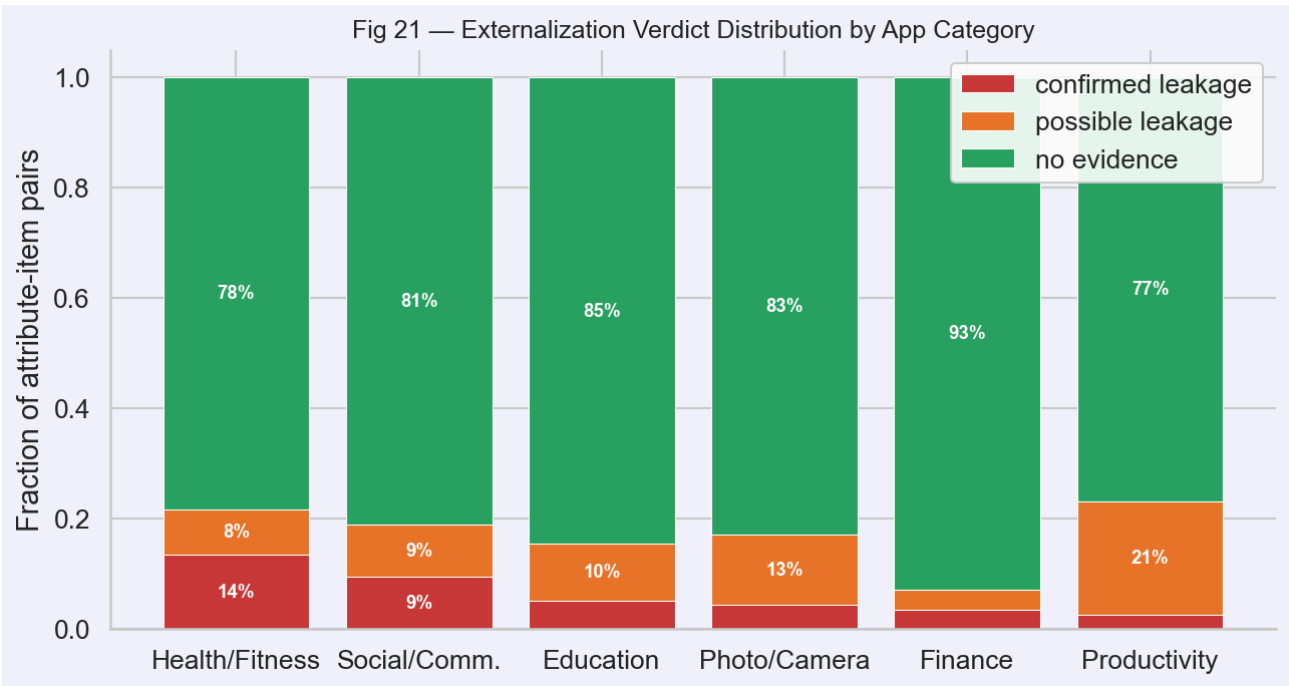


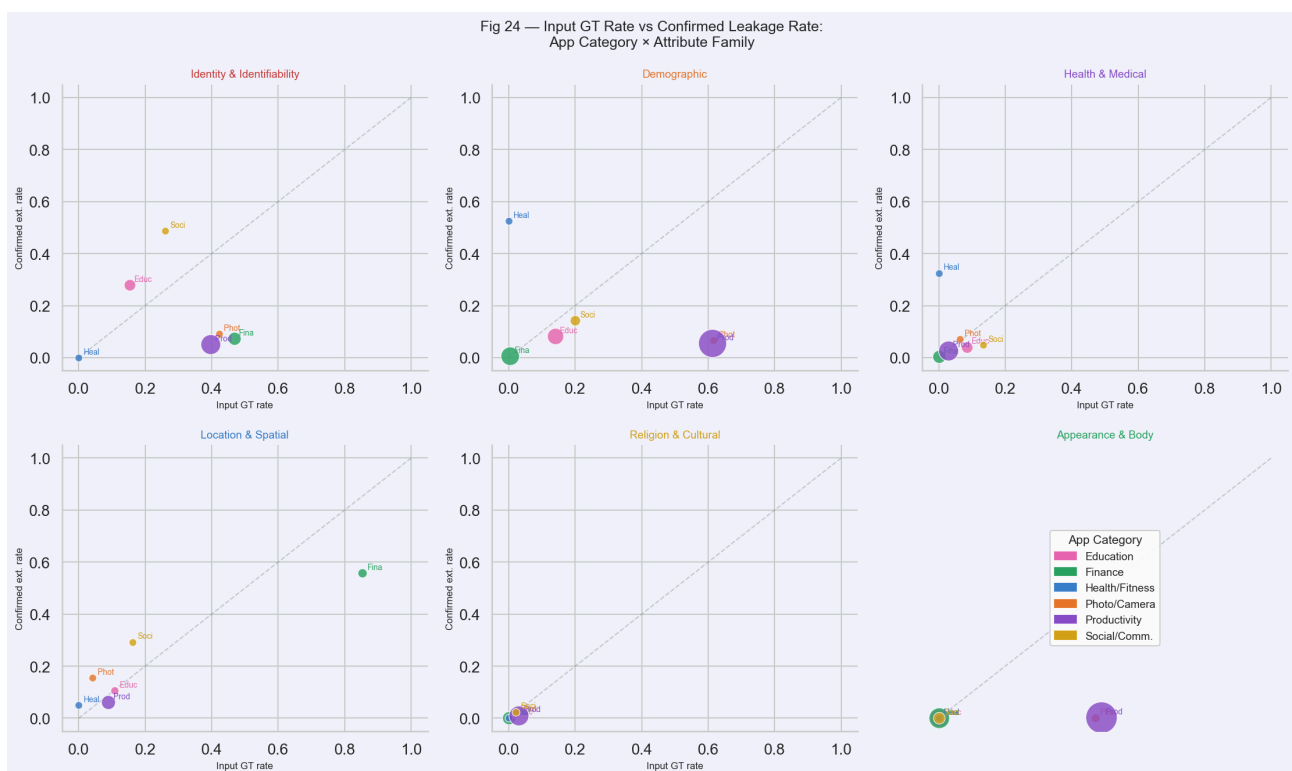
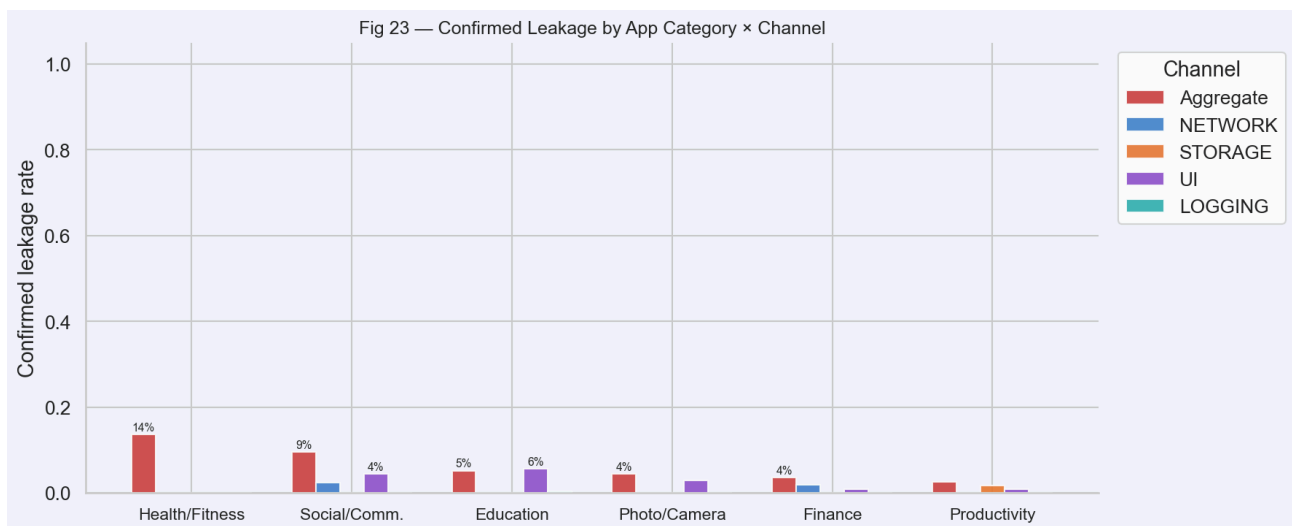
6b. Leakage by App Category

Apps are grouped into six functional categories: **Finance**, **Photo/Camera**, **Productivity**, **Education**, **Social/Communication**, and **Health/Fitness**.

- **Health/Fitness** (20.0 items): 13.6% confirmed, 21.7% any leakage
- **Social/Comm.** (106.0 items): 9.5% confirmed, 18.9% any leakage
- **Education** (292.0 items): 5.2% confirmed, 15.4% any leakage
- **Photo/Camera** (70.0 items): 4.4% confirmed, 17.2% any leakage

- **Finance** (388.0 items): 3.5% confirmed, 7.1% any leakage
- **Productivity** (852.0 items): 2.6% confirmed, 23.2% any leakage





7. Recommendations

1. **Prioritize identity and location filtering** across all app types — these attributes leak at the highest rate regardless of input modality.
2. **Audit STORAGE channels** (databases, memory systems, on-device logs) — STORAGE leakage is non-visible to users and persists beyond session boundaries.
3. **Deploy inference-aware output filters for image→text apps** — the text description of an image is a semantic consolidation point for many visual attributes that were diffuse in the original image.
4. **Test for inference expansion** — apps should be evaluated not just for passthrough of GT-present attributes, but for attributes that appear in the output that were NOT in the input (implicit inference).

5. **Treat multi-attribute leakage as the norm** — privacy defenses that target individual attribute leakage miss the broader threat of simultaneous multi-attribute profile construction.

Appendix

A. Per-App Detailed Statistics

App	N pairs	Confirmed %	Any Leak %
healyks	420	13.6%	21.7%
waico	1,764	10.8%	21.3%
tool-neuron	231	9.1%	27.7%
deeptutor	4,725	6.9%	15.0%
llm-vtuber	819	6.7%	13.7%
chat-driven-expense-tracker	2,394	4.6%	11.2%
google-ai-edge-gallery	1,260	3.6%	15.2%
budget-lens	5,754	3.1%	5.5%
snapdo	16,569	2.6%	25.1%
pocketpal-ai	2,289	2.3%	9.4%
edupal	1,659	0.2%	16.7%
spendsense	21	0.0%	4.8%

B. Per-Dataset Sample Counts

Dataset	Input Type	N pairs	Confirmed %
ASAP-AES	text	2,436	0.2%
GretelSyntheticPII	text	2,394	4.6%
HR-VISPR	image	17,640	2.7%
MIMIC-CXR	image	210	3.8%
MultiCaRe	text	630	14.6%
OpenPII	text	210	2.9%
PrivacyLens	text	6,951	8.5%
SROIE2019	docs	5,775	3.1%
SynthPAI	text	1,659	0.7%

C. Attribute Family → Member Mapping

Family	Attributes
Identity & Identifiability	face, identity
Demographic	age, gender, race, marital status
Health & Medical	disability, medical

Location & Spatial	location
Religion & Cultural	religion, ethnic_clothing
Appearance & Body	nudity, height, weight, haircolor, color
Attire, Role & Group	formal, casual, uniforms, troupe
Activity & Lifestyle	sports

D. Data Quality Notes

- prompt4/5 items with failed `ext_eval` (no verdict) are excluded from all analyses.
- `spendsense|SR0IE2019` (n=1) and `tool-neuron|HR-VISPR` image→image (n=1) excluded as statistically insignificant.
- `output_eval` (raw app output stage) is only available for PrivacyLens text→text runs (deeptutor, waico, llm-vtuber, tool-neuron).
- Channel-level statistics are conditioned on the channel being present in that item's externalization record.